OUR PROTOTYPE

Straps that lock like a zip tie



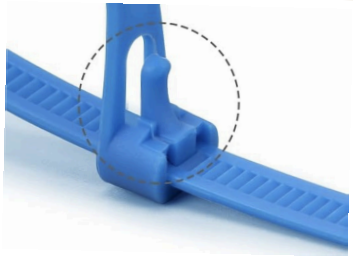
Front



Side



Back



The reusable zip tie behind our closure

Printed straps wrap the ankle in the athletic-tape X and tighten with one hand.

What we tested

PETG with holes broke when bent	NO
TPU with holes layers split	NO
Solid TPU firm, moulds slightly	CHOSEN
Straps at key points ankle stayed still	CHOSEN

LEARN MORE

Scan to see our work

Website

CAD, testing and videos

Engineering notebook

Our D.A.A.N. journal

MEET THE TEAM

Gertrudis · Jose · Nancy

Teachers: Lupita C. Vasquez and Claudia Cortez

Gladys Porter Early College High School
Brownsville, Texas

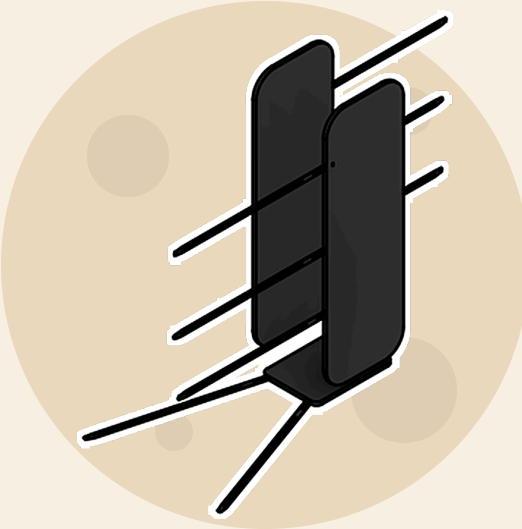
A student design project. Not a medical device.

LUNOVA

NASA HUNCH · LUNAR SPLINT DESIGN CHALLENGE

OrthoPrint

A custom-fit ankle splint, 3D-printed on the Moon.



Team Lunova · Gertrudis, Jose & Nancy
Gladys Porter Early College High School
Brownsville, Texas

THE PROBLEM

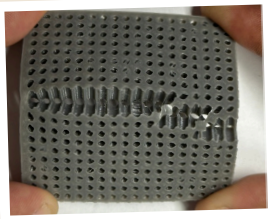
No hospital on the Moon

If an astronaut tears an ankle ligament on a lunar base, the ride home takes days. The crew needs a splint that fits, made right there.

$1,000 to launch every gram, so a box of splints in every size is out

1/6 g lunar gravity, and weeks of wear in a closed habitat

3–26 weeks for a torn ankle ligament to heal



Holes split, so we went solid



Hanging-weight test

WHAT WE HEARD

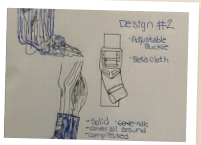
Needs Assessment Survey · 56 students

Ankle injured most	63%
Hard to find a fit	54%
On their feet a lot	96%
Devices help healing	98%

HOW IT WORKS

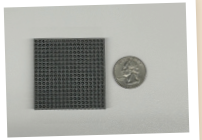
From scan to splint

1 Scan
Scan the injured astronaut's lower leg.



Early concepts

2 Print
Print the shell and straps on the base.



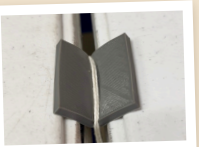
TPU sample

3 Fit
Unlatch it and slide it on from above.

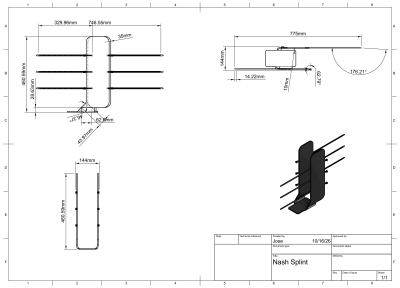


Our CAD

4 Strap
Wrap the straps in an X and tighten with one hand.



Strap load test



Detail drawing, dimensions in mm · Jose

LUNAR INNOVATIONS

Built for the Moon

1 Single-piece ratchet closure
Tightens or releases with one hand. No loose parts to lose.

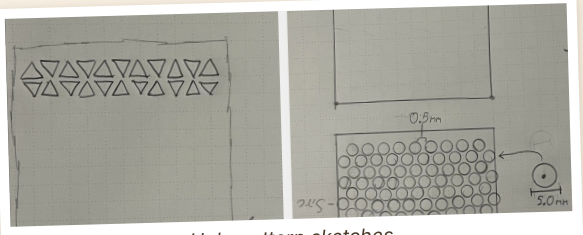
2 Integrated hygiene
Comes off to clean the splint and check the skin.

3 Thermal phase design
Flexible to put on, firm to hold the ankle still.

MATERIALS

PLA · ABS · TPU · PETG

CF-PEEK for flight



Hole pattern sketches

Next steps

- Test strap thickness and break load
- Find where the filament softens with heat
- Build prototype #2 for the CDR